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Before the

U.S. COPYRIGHT OFFICE
LIBRARY OF CONGRESS

In the Matter of Software-Enabled Consumer Products Study

Docket No. 2015-6

Reply Comments of Engine Advocacy

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Table of Contents

Engine Advocacy ("Engine") submits these reply comments in response to the Copyright Office's Notice of Inquiry and Request for Public Comment for its Software-Enabled Consumer Products Study.

Engine supports the growth of technology entrepreneurship through economic research, policy analysis, and advocacy on local and national issues. Engine works with the White House, Congress, federal agencies, state and local governments, and also international advocacy organizations to educate and inform them of the changing face of American high-tech entrepreneurship and issues vital to fostering technological innovation.

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I. Introduction

The landscape of software-enabled consumer products holds vast opportunities for innovation, interoperability, increased consumer choice and utility, and enhanced social value. These opportunities can best be realized to their full potential by interpretations of and modifications to copyright law that facilitate innovation and avoid tensions between the law and this rapidly developing field.

Engine's initial comments emphasized in part how allowing copyright to lock down technology necessary for interoperability—specifically, software such as application programming interfaces ("APIs")—would frustrate innovation and competition. The inability of users to modify the software in devices they own hampers innovation, prevents interoperability, reduces security, and undermines settled expectations of ownership. Ultimately, Engine encouraged the Copyright Office to recommend changes that secure the open use of functional software to facilitate interoperability and ensure that users are treated as owners of the software in their devices and can tinker with, improve, or repair those devices by analyzing, copying, and modifying the software.

In this reply, Engine strives to address three recurrent arguments, raised by some parties in their initial comments, that are important to the relationship between copyright law and software-enabled consumer products. First, software-enabled consumer products possess unique characteristics that could allow copyright restrictions to over-burden legitimate user interests. Second, the fact that software is an innovative industry should not mean that improvements to copyright law won't have the potential to lead to additional innovation. Third, the prudent approach at this stage is for the Copyright Office to move proactively and decisively to ensure that copyright law best facilitates innovation in embedded software, rather than waiting until innovation has been chilled.

II. Copyright Law Has a Particular Impact on Embedded Software.

Some commenters have asserted that it is "virtually impossible to draw any principled distinction between embedded and non-embedded software."[1] While it is true that embedded and standalone software possess a dynamic and entwined relationship with one another, embedded software has important distinct qualities from standalone software. While infringement is less of a threat for embedded software than it is for standalone, copyright restrictions in embedded software are particularly burdensome it can limit essential interoperability is essential and disrupt consumers' settled expectations regarding their ownership rights in physical products.[2]

As Public Knowledge and New America Foundation's Open Technology Institute noted in their comments, "embedded software does not raise the same concerns about infringement that have driven software copyright policy in other areas."[3] Congress itself has acknowledged this distinction and has applied it to existing copyright law, demonstrating not only the relevance of the distinction but also that it is possible to legally recognize it.[4] Copying embedded software for resale generally provides little value, and so there is incentive to do so. Unlike standalone software, which can be vulnerable to piracy because it can run on any compatible computer platform and consumers often already own the platform, embedded software typically operates only with the accompanying hardware in which it is embedded and has no separate use or purpose. Moreover, concerns that competitors might copy the software in software-enabled products for their own devices, essentially free-riding on others' development work, are misplaced. Copying or pirating software for such replacement purposes should remain an unlawful infringement of copyright even if repair and interoperability are properly recognized as non-infringing activities.[5] As an owner of a book "is entitled, without the authority of the copyright owner, to sell or otherwise dispose of the possession of that copy" after purchasing, so should an owner of a software-enabled consumer product be entitled. Similarly, as an owner of a book cannot generally lawfully copy the text of that book to republish or distribute it, neither should an owner of a software-enabled consumer product lawfully copy the embedded software and duplicate it within another, competing product.[6]

Because embedded software raises minimal infringement concerns, over-restrictive copyright protections serve primarily to limit users' legitimate interests rather than protect against illegal activity. One of the most important legitimate interests is promoting interoperability. As the "Internet of Things" proliferates, software-enabled consumer products will increasingly need to interoperate with platforms, networks, and devices made by diverse companies. Machine-to-machine communication is fundamental to the value of these products;

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[1] BSA|The Software Alliance, *Comments of BSA /The Software Alliance*, 2. Other commenters that advocated similarly include: Microsoft Corporation, Owner's Rights Initiative, Software and Information Industry Association, ACT|The App Association, Global IP Center, Entertainment Software Association, Center for Democracy and Technology, and the Computers and Communications Industry Association.

[2] Infringement is less of a threat both in the reduced likelihood that individuals will infringe and in that any infringement is less damaging to the market.

[3] Public Knowledge and Open Technology Institute, *Comments of Public Knowledge and Open Technology Institute* 2.

[4] 17 U.S.C. § 109 declares that software rental limitations do not apply to a "computer program embodied in a machine or product and which cannot be copied during the ordinary use of the machine or product."

[5] 17 U.S.C. § 102.

[6] 17 U.S.C. § 109.

the inability to interoperate freely would greatly reduce consumer choice and innovation. Permitting exclusive rights on the ability to interoperate will permit software rights-holders to become gatekeepers to products, industries, and innovation in ways we would never countenance for other property, while providing few if any benefits in protecting against unlikely infringement. Interoperability also can be important for standalone software, but the lack of interoperability is particularly devastating to software-enabled products.

Embedded software is distinguishable from standalone software by its very form: it is embedded into a physical consumer product and not sold separately. This distinction lends itself to a unique relationship with consumers. Decades of traditional ownership rights have instilled strong expectations about the relationship between owners and their products, which the law should continue to respect. The traditional rights that have always accompanied physical objects, including the right to tinker, repair, lend, and transfer, should extend as well to the embedded software within these objects. As detailed in Engine's initial comments, without these parallel rights, innovation, safety, privacy, consumer choice, and more will be frustrated.

Additionally, inappropriate application of software law to embedded software makes such software particularly vulnerable to new and unwarranted restraints on the alienability of property. Unless users preserve their traditional expectations and rights of use, manufacturers will sell their software-enabled consumer products accompanied by a copyright license in the embedded software and claim that the user therefore does not own the entire product. This would imperil the ability of a consumer to resell their used car or repair their refrigerator—behaviors consumers reasonably expect to be lawful—without violating copyright law. Even the current uncertainty that consumers face surrounding their ownership rights is damaging, as basic property rights are publicly called into question or explicitly repudiated by manufacturers.

Ultimately, copyright restrictions have a disproportionate impact on embedded software. Balancing the costs and benefits of different approaches to applying copyright to embedded software suggests that over-enforcement will lead to significant barriers to innovation, interoperability, competition, and safety with little benefit to consumers or the incentive to innovate in the first place. The Copyright Office and Congress should be vigilant to prevent the growth of such barriers and examine carefully issues of copyrightability, fair use and the first sale doctrine as applied to embedded software.[7]

III. The Existence of Strong Innovation Under the Current Copyright Regime Does Not Mean that an Improved Copyright System Would Fail to Spur More Innovation.

The fact that the technology industry has boomed under the existing legal framework does not mean that framework is optimal for innovation or business in embedded software. The status quo of the software industry does not tell us everything we need to know about the

[7] While ownership rights and interoperability are of unique importance to embedded software, Engine does not intend to suggest that non-embedded, standalone software does not need and deserve the same freedom from copyright-based restrictions on interoperability, alienability, or other socially beneficial, fair uses. The question of how these issues should be treated for non-embedded software is outside the scope of the Notice of Inquiry and is therefore not addressed here.

counterfactual—namely, the innovations that might otherwise have occurred under a different set of rules. While the current copyright regime has undoubtedly contributed to some extent to the current level of innovation, adapting appropriately to the market going forward can continue to improve the amount and pace of future innovation.[8]

Various commenters have argued that, for example, "strong copyright protection for software (embedded and otherwise) has been a tremendous policy success that has enabled decades of innovation and creativity," and therefore embedded software, governed by the current system, will continue to be successful, obviating the need for any systemic changes.[9] While this argument is facially plausible, a deeper look reveals that it is grounded on logical fallacies and rests on the incorrect notion that some innovation under current copyright law rules out the possibility of increased innovation under an improved system.

First, there is the deductive error of "affirming the consequent," where one assumes that an effect follows one potential cause because it happened after that cause. For example, it is common sense that if it rains, the street will be wet. To affirm the consequent is to conclude that because the street is wet, it must have rained. This is, of course, incorrect. It is likewise incorrect to conclude that current copyright law caused innovation solely because that innovation happened. Good copyright law may make the software industry more successful. But the success of software industry does not mean that copyright law is wholly good; many factors have contributed to the massive innovation we have seen in the field of software. In fact, copyright law could have had an adverse impact on innovation that was offset by other positive factors, and still resulted in the innovation we have seen.

Second, the argument relies on weak inductive reasoning. Unlike deductive reasoning, inductive reasoning produces conclusions that are inherently uncertain but can be strong or weak based on the evidence given. For example, strong inductive reasoning suggests that because, every day to date the law of gravity has held, it will probably hold tomorrow. In this proceeding, some commenters have argued that since software, governed by copyright law, has been successful, it will continue to be successful under existing interpretations and application of copyright law. This is far from strong inductive reasoning. The relevant copyright provisions have governed software for only forty years—essentially the entire relevant existence of "software"—and the nature of the software industry changes more year-to-year than virtually any other market segment. Because of the unique characteristics of embedded software, described above, the history and nature of innovation in software in general does not necessarily mean that the same conditions will govern the coming future of ubiquitous, interoperable embedded software in consumer devices. In any event, even if software as an overall industry might continue to flourish—which is far from certain—innovation in the specific context of embedded software might be restrained.

While it is difficult to be sure how the world would have developed under a different legal regime, the back-and-forth in the online music industry demonstrates the likely presence

[8] Commenters that attribute the success of the software industry to copyright law include: Microsoft, Entertainment Software Association, BSA, Global IP center, Copyright Alliance, Software and Information Industry Association.
[9] The Entertainment Software Association, *Comments of the Entertainment Software Association*, 2.

and scale of missed opportunities in this constantly changing industry: a decade-long delay in a billion-dollar market. In 2012, Michael Carrier conducted anonymous interviews of 31 “CEOs, company founders, and vice presidents from technology companies, the recording industry, and venture capital firms” regarding their “reactions to the district court’s injunction in the case involving peer-to-peer (p2p) service Napster.”[10] After lawsuits by A&M Records and 17 other record companies, both the district court and the Ninth Circuit held that Napster was liable for direct and contributory infringement. Whatever the merits of that liability finding, on a policy level, the *Napster* decision had major repercussions on the development of online music for years afterward.

In retrospect, Carrier argues, even the winners did not benefit from the copyright decisions in the long run. One executive of a major record label lamented on the “missed opportunity” and observed that “it made ‘no sense’ that ‘fifteen years into digital exploration…when I buy a song, I still only get that song’” while the labels could offer “additional extras, like exclusive interviews and backstage access.”[11] Another compared the industry with the booming market of e-books, where publishers and retailers had seized on and included in the bundle innovative services such as “relationship with the author” and “social reading environment.”[12] Because the recording industry had adopted a strategy that relied primarily on the application of copyright law, it failed to fully recognize or engage with an inevitable trend—online music.

One respondent of the Carrier survey speculated that “if the court had come out the other way, ‘every television broadcast, every piece of music ever created, [and] every image ever taken’ could be ‘available unfettered to all of the devices we have.’”[13] She concluded that “if Napster had won, I guarantee you, it would be a $50 billion market right now.”[14] Just one week after her interview with Carrier, Spotify launched in the United States.[15] Sean Parker, co-founder of Napster, invested in Spotify in 2010 and had been a board member since then. In June 2015, the company valued at $8.5 billion.[16] Although Parker described Spotify as “actually very similar to what I dreamt of 10 years ago,” the question remains what might have happened to innovation, to the industry and to the benefit of consumers if Parker had not had to start over again.

IV. The Risks to Innovation of Applying Current Copyright Law in the Fast-Moving, Emerging World of Software-Enabled Consumer Devices Warrant Moving Quickly and Deliberately to Make Appropriate Changes to the Law.

Some commenters in the initial round suggested that it is “premature to recommend legislative changes” at this time and urged the Copyright Office to “proceed with caution.” This

[10] Michael Carrier, *Copyright and Innovation: the Untold Story*, WIS. L. REV. 891, 891 (2012).
[11] *Id.* at 952.
[12] *Id.*
[13] *Id*. at 909.
[14] *Id.*
[15] The interview happened on Nov. 23, 2011. See *supra*, at 908, n. 106. Spotify launched in the United States on Nov. 30, 2011.
[16] Kasper Viita & Matthew Campbell, *Spotify Value Tops $8 Billion as Investors Bet on Streaming*, http://www.bloomberg.com/news/articles/2015-06-10/spotify-valued-at-8-2-billion-as-teliasonera-buys-stake.

approach is backwards.[17] As some of these same commenters acknowledged, “we are in the midst of a transformational new generation of software innovation” accompanied by a “rapid pace of innovation.”[18] But the lessons they suggest taking away from this rapid transformation are wrong. This rapidly developing innovation is best facilitated and protected *not* by waiting to see what issues develop and how much the barriers that copyright law impose interfere, but rather by engaging the limitations and weaknesses of copyright law as soon as they are recognized and making changes needed to allow this innovation to flourish without undue interference or restriction.

The negative consequences of undue hesitation and delay are especially great in a field where dynamic change is ever-present and accelerating. Take instant messaging software, for instance. When MSN Messenger was first launched in 1999, it included only basic functions such as plain text messaging and a contact list. Today, WeChat, one of the most popular instant messenger programs, incorporates texting, emoticons, voice messaging, video conferencing, file sharing, and even money transfers. While “tomorrow” may be more or less similar to “today” in many fields, software is assuredly not one of them, and it would be a mistake to regulate as if it were.

A powerful example of the value ensuring that the law is calibrated proactively to ensure opportunities for innovation is provided by § 230 of the Communications Decency Act.[19] Section 230 is perhaps Congress’s greatest success in getting out in front of a rapidly developing technology—rather than sitting back and waiting for problems to arise—in a way that paved the way for tremendous subsequent innovation. In 1996, recognizing the critical importance of promoting “the continued development of the Internet . . . and other interactive media,” and preserving the vibrant and competitive free market that presently exists for the Internet and other interactive computer services,”[20] Congress enacted § 230. Section 230 ensures that “[n]o provider or user of an interactive computer service shall be treated as the publisher or speaker of any information provided by another information provider.”[21] Courts have construed the statute “broadly to extend protection from liability to many companies.”[22]

Prior to § 230, in the early days of the Internet, a company could be held liable for any defamatory or other tortious content that appeared on its website, regardless of who wrote it. Section 230 established the distinction between a publisher/speaker and a provider/user, thus significantly reducing the risk of liability that Internet services face for the user-generated content they host. As a result, for instance, Facebook cannot be held liable for the defamatory content posted by any of its 1.6 billion users (though, of course, users themselves may still be

[17] BSA | The Software Alliance, *Comments of BSA /The Software Alliance*, 5. Microsoft Corporation, *Comments of Microsoft Corporation*, 8..
[18] Microsoft Corporation, *Comments of Microsoft Corporation*, 2.BSA The Software Alliance, *Comments of BSA The Software Alliance*, 1.
[19] 47 U.S.C. § 230(c)(1).
[20] 47 U.S.C. § 230(b)(2).
[21] *Id*.
[22] Cyrus Sarosh Jan Manekshaw, *Liability of ISPs: Immunity from Liability under the Digital Millennium Copyright Act and the Communications Decency Act*, 10 COMP. L. REV. & TECH J. 101, 112 (2005).

held liable).[23] Without § 230, the number of users and the volume of content they generate would create the potential for massive liability and no company, particularly no startup, could withstand the risk and potential expense of litigating and settling large numbers of lawsuits. Litigation can easily drag on for years and impose legal fees that would quickly bankrupt a start-up, even if it ultimately prevails on the merits.[24] It takes just *one* suit, no matter the result, to bring many typical start-ups to a halt. An all too common analogy in recent years has been the conflicts between patent assertion entities ("patent trolls") and start-ups. Knowing that start-ups typically cannot afford the legal fees necessary to get spurious claims dismissed, patent trolls survive by filing questionable lawsuits on weak patents and insisting on extortionate settlements.

Section 230 and its safe harbor are widely recognized as critical elements in the remarkable flourishing of innovation that has occurred since 1996 in all aspects of Internet technology involving user-generated content. Without the safe harbor § 230 created, the chilling effect of potential litigation on innovation and start-ups would have been enormous and enormously damaging. When § 230 was passed, companies that depend on user-generated content such as eBay (1995) and Craigslist (1995) were start-ups. Policy-makers could not have foreseen the social media revolution that would later be brought by the likes of YouTube (2005), Facebook (2004), and Reddit (2005). These companies and the profound innovations they embody were either nascent or nonexistent at the time, and many of them would either not exist or exist in very different form without the § 230 safe harbor.[25] Professor Eric Goldman attributes "a non-trivial role" to § 230 "in the ascension of Internet as a dominant media."[26] Anupam Chander describes the "shield" provided by § 230 as "invaluable."[27] Derek Kanna explains that it "cleared the way for the modern Internet" and was "the law that gave us websites like Reddit, Craigslist, Digg, and perhaps all of social media."[28] Section 230 has been "economically powerful, perhaps in ways Congress couldn't have foreseen. . . . As a result, it has functioned as a permission slip for the whole Internet that says: 'Go innovate.'"[29] Without § 230, the Internet would look very different—and far less innovative—today. In 1996, the Web had a total number of 257,601 sites—a negligible number compared with 1 billion today, yet the lawmakers made the correct choice then to act promptly.[30]

Ironically, supporters of the status quo before § 230 might well have made similar arguments that any change in the law should await the development of significant problems, and the innovations that would rebut their case would never have appeared since they would never

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[23] Number of Monthly Active Facebook Users Worldwide as of 4th Quarter 2015, http://www.statista.com/statistics/264810/number-of-monthly-active-facebook-users-worldwide/.
[24] Carrier, 972.
[25] *See The State of Broadband 2014: Broadband for All*, Broadband Comm'n for Digital Dev. [UN agency for information and communications technology] (Sept. 2014), *available at* http://www.broadbandcommission.org/documents/reports/bb-annualreport2014.pdf.
[26] Eric Goldman, *Speech Shutdown at the Virtual Corral*, SANTA CLARA COMPUTER & HIGH TECH. L.J. 101, 109 (2005).
[27] Anupam Chander, *How Law Made Silicon Valley*, 63 EMORY L.J. 639, 653 (2004).
[28] Derek Khanna, *The Law that Gave Us the Modern Internet—and the Campaign to Kill It*, The Atlantic (Sept. 12, 2013), *available at* http://www.theatlantic.com/business/archive/2013/09/the-law-that-gave-us-the-modern-internet-and-the-campaign-to-kill-it/279588/.
[29] *Id*.
[30] Total Number of Websites, http://www.internetlivestats.com/total-number-of-websites/.

have been founded or funded in the face of potentially crippling liability. Fortunately, rather than sitting back and waiting until the failures of the status quo became self-evident, Congress actively paved the way for innovation and progress in the "rapidly developing array of Internet and other interactive computer services available to individual Americans."[31] The Copyright Office should adopt this same principle as it seeks to facilitate and preserve this next wave of economically valuable, rapidly developing innovation.

Commenters' also attempt to rely on language in the Department of Commerce's Internet Policy Task Force's ("IPTF") *White Paper on Remixes, First Sale, and Statutory Damages*, but that reliance is misplaced.[32] The White Paper notes that "the alienability of everyday functional products is an important issue for consumers" and "further attention would be warranted" if the market develops so that this is an issue.[33] Settled expectations about ownership and alienability are indeed important issues for consumers, but developments in the market show the time already is ripe for appropriate copyright reform. For example, in 2015, John Deere and General Motors advanced a view of embedded software that challenges the very idea of owning a tractor or a car: the purchaser of a vehicle does not fully own all of it—rather, she "receives an implied license for the life of the vehicle to operate the vehicle."[34] This indicates that embedded software is already creating challenges for expectations of alienability, justifying action from the Copyright Office and Congress now, not down the road when the problems have become worse or innovation has been chilled. Otherwise, the uncertainty surrounding the law will stifle invention and creativity and startups and markets will be restrained.

Ultimately, waiting to modernize copyright law or approaching changes with a paralyzing level of caution or hesitation is unwise and unwarranted. We are past the dawn of software-enabled consumer products—consumer software is found in many of our everyday products—and soon it will become truly ubiquitous. Examples abound. Just last week, the New York Times described how app makers and "makers of digital home devices like Nest are also rushing to make their products compatible with" Amazon's Echo.[35] Earlier this month, Wired featured an article titled *How to Link Your Car's Brain to Your Smart Home*, detailing a $100 dollar device that enables your car to unlock the front door to your home, turn on your house lights, and open your garage automatically by turning off the car in your driveway.[36] The integration of the Internet of Things into everyday life is a reality, not a hypothetical,[37] and the law should not be allowed to lag behind. In any event, beginning to take steps toward needed reforms today will not result in immediate changes since any recommendations from the Copyright Office will take

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[31] 47 U.S.C. § 230(a)(1).

[32] Commenters include Microsoft, the Software and Information Industry Association Commenters, BSA | The Software Alliance, and ACT | The App Association.

[33] Dep't of Commerce, Internet Policy Task Force, *White Paper on Remixes, First Sale, and Statutory Damages*, 64 (2016), *at* http://www.uspto.gov/sites/default/files/documents/copyrightwhitepaper.pdf.

[34] 16 John Deere Reply Comment, 2015 DMCA proceeding, http://copyright.gov/1201/2015/comments-032715/class%2021/John_Deere_Class21_1201_2014.pdf at 6.

[35] Farhad Manjoo, *The Echo From Amazon Brims With Groundbreaking Promise*, The New York Times (Mar. 8, 2016), *available at* http://www.nytimes.com/2016/03/10/technology/the-echo-from-amazon-brims-with-groundbreaking-promise.html?action=click&contentCollection=Sports&module=MostPopularFB&version=Full&region=Marginalia&src=me&pgtype=article.

[36] Michael McCole, *How to Link Your Car's Brain to Your Smart Home*, Wired (Mar. 3, 2016), *available at* http://www.wired.com/2016/03/iot-cookbook-automatic/.

[37] *See* The Entertainment Software Association, *Comments of the Entertainment Software Association*, 7.

time to be enacted. Accordingly, the Copyright Office should engage in needed reexamination and reform of the law affecting these innovative technologies without delay and without waiting for examples of harm to innovation.

V. Conclusion

Engine reiterates its appreciation for the Copyright Office's current study of software-enabled products. Based on the comments submitted in the first round of this NOI, Engine urges the Office to recommend and strongly support changes to and interpretations of copyright law and policy that maximize the tremendous opportunities for innovators, entrepreneurs, startups, and users that have been and will be created by the spread of software-enabled products and the Internet of Things. The Office should not accede to invitations to do nothing now and wait to act until problems have become obvious and critical. The better course is to act now to pursue near-term changes and clarifications that will help ensure, before it is too late, that copyright law does not restrict the next generations of innovative new products and services. Protecting innovation and its benefits is too significant to leave for another day; the Office should support necessary legislative changes that proactively provide the best possible conditions to ensure the opportunities for innovation continue to grow.